Complete 6-Month Intensive Java Developer Roadmap: From Zero to Job-Ready

For someone starting with basic Java programming knowledge and committing 8 hours daily, this roadmap provides a structured, day-by-day plan with specific topics, daily routines, DSA practice scheduling, and project milestones to secure a Java developer job within 6 months. [1] [2] [3]

PHASE 1: CORE JAVA FUNDAMENTALS (Weeks 1-8 | Days 1-56)

Time Allocation Per Day: 8 Hours

- **2-2.5 hours**: Learning concepts (videos, documentation, blogs)
- **2-2.5 hours**: DSA practice (competitive coding problems)
- **2-2.5 hours**: Core Java hands-on coding projects
- **1-1.5 hours**: Review, debugging, documentation writing

Week 1: Java Basics & Programming Fundamentals (Days 1-7)

Daily Breakdown:

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 1	JDK/JRE setup, Variables, Data Types, Type Casting	2	0.5	1.5	Environment setup, primitive vs reference types
Day 2	Input/Output (Scanner class), Comments, Naming conventions	1.5	0.5	2	Basic I/O programs
Day 3	Operators (Arithmetic, Logical, Relational, Bitwise)	2	1	1.5	Operator precedence, instanceof
Day 4	Control Flow (if-else, switch-case, ternary operator)	2	1	1.5	Nested conditionals
Day 5	Loops (for, while, do-while, for-each, break, continue)	2	1.5	1	Nested loops, loop optimization
Day 6	Practice Day - Build mini console programs	1	1.5	3	Calculator, number patterns, temperature converter

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 7	Review & Concepts Consolidation	1	1.5	2.5	Q&A, debugging common issues

Week 1 DSA Focus: Start with **basic array problems** on LeetCode/HackerRank

- Problem types: Find max/min, reverse array, two sum easy variant
- Target: 5-7 problems solved (Easy difficulty)
- Time per problem: 15-20 minutes

Week 1 Core Java Project: Build a **Simple Console Banking Menu System**

```
// Features: Display menu, take input, show balance, deposit/withdraw
// Demonstrates: Variables, input/output, conditionals, loops
```

Week 2: Arrays, Strings & Basic Collections (Days 8-14)

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 8	Arrays (declaration, initialization, multidimensional)	2	1.5	1.5	Array indexing, iteration
Day 9	String handling (String, StringBuffer, StringBuilder)	2	1.5	1.5	String immutability, methods
Day 10	String manipulation & important methods	1.5	1.5	2	substring, replace, split, indexOf
Day 11	Java Collections (ArrayList, HashMap basics)	2	1.5	1.5	Dynamic arrays, hash tables intro
Day 12	More Collections (HashSet, LinkedList, Queue)	2	1.5	1.5	Collection types, use cases
Day 13	Practice Day - Array & String problems	1	2	3	Sorting, searching, string rotation
Day 14	Review & Collections Deep Dive	1	1.5	2.5	Internal implementation basics

Week 2 DSA Focus: Arrays & Strings

- Problem types: Two pointers (reverse string, merge sorted), prefix sums, string pattern matching
- Target: 10-12 problems solved
- Recommended patterns: Linear search, binary search variants, palindrome checking

Week 2 Core Java Project: Build a **Student Grade Management System**

```
// Features: Add students with marks, calculate average, sort by grade
// Demonstrates: ArrayList, Arrays, String parsing, Collections usage
```

Week 3: Object-Oriented Programming Fundamentals (Days 15-21)

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 15	Classes & Objects, State & Behavior, Instance vs Static	2.5	1	1.5	Class structure, object instantiation
Day 16	Constructors (Default, Parameterized, Constructor overloading)	2	1	1.5	Constructor chaining, this keyword
Day 17	Methods (Instance, Static, Method overloading)	2	1	1.5	Method signatures, parameter passing
Day 18	Access Modifiers (public, private, protected, default)	1.5	1.5	2	Encapsulation, data hiding
Day 19	Getters/Setters, Immutable Objects	2	1	1.5	Object state management
Day 20	Practice Day - OOP fundamentals projects	1	1.5	3	Bank account, employee records
Day 21	Review & OOP Principles	1	1.5	2.5	Design patterns basics

Week 3 DSA Focus: Continue with **array/string problems** while learning OOP

- Problem types: Matrices, sliding window algorithms, frequency counting
- Target: 10-12 problems solved
- Introduce: Time & space complexity analysis (Big O notation)

Week 3 Core Java Project: Build a **Banking System with Classes**

```
// Classes: Account, Customer, Transaction
// Features: Multiple accounts, transaction history, balance inquiry
// Demonstrates: Classes, constructors, encapsulation, methods
```

Week 4: Advanced OOP - Inheritance & Polymorphism (Days 22-28)

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 22	Inheritance (extends keyword, super keyword, IS-A relationship)	2.5	1	1.5	Parent-child classes, method overriding
Day 23	Method Overriding, super() in constructors	2	1	1.5	Dynamic method dispatch
Day 24	Polymorphism (Compile-time & Runtime), instanceof	2	1	1.5	Late binding, method resolution
Day 25	Abstraction (abstract classes, abstract methods)	2	1.5	1.5	Abstract class design, interfaces intro
Day 26	Interfaces (multiple inheritance, contract definition)	2	1.5	1.5	Interface implementation, default methods
Day 27	Practice Day - Inheritance & Polymorphism projects	1	1.5	3	Shape hierarchy, vehicle system
Day 28	Review & OOP Design Principles	1	1.5	2.5	SOLID basics

Week 4 DSA Focus: Linked Lists Introduction

- Topics: Node creation, insertion, deletion, traversal, reversal
- Problem types: Detect cycle, merge linked lists, find middle
- Target: 10-12 problems solved
- Complexity: Move towards Medium difficulty problems

Week 4 Core Java Project: Build a **Employee Management Hierarchy**

```
// Classes: Employee (abstract), Manager, Developer, Designer
// Features: Polymorphic salary calculation, role-based operations
// Demonstrates: Inheritance, abstraction, polymorphism, interfaces
```

Week 5: Exception Handling & Advanced Collections (Days 29-35)

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 29	Exception Handling (try-catch-finally, throw, throws)	2	1	1.5	Exception hierarchy, error vs exception
Day 30	Custom Exceptions, Multiple catch blocks	2	1	1.5	Exception propagation, best practices

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 31	Java Collections Deep Dive (ArrayList, HashMap, HashSet)	2	1.5	1.5	Internal implementation, complexity analysis
Day 32	Generics in Java (Type parameters, wildcards, bounds)	2	1	1.5	Type safety, generic classes & methods
Day 33	Collections Utility methods, Comparable & Comparator	2	1.5	1.5	Sorting custom objects, Collections.sort()
Day 34	Practice Day - Exception handling & Collections	1	1.5	3	Custom exception projects, sorting algorithms
Day 35	Review & Collections Framework Mastery	1	1.5	2.5	Interview Q&A on collections

Week 5 DSA Focus: Stacks & Queues

- Topics: LIFO/FIFO principle, implementation using arrays/linked lists
- Problem types: Balanced parentheses, next greater element, sliding window max
- Target: 10-12 problems solved
- Important: Understand when to use stacks/queues vs other data structures

Week 5 Core Java Project: Build a **Library Management System**

```
// Classes: Book, Member, Librarian, Library
// Features: Add/remove books, member registration, issue/return tracking
// Demonstrates: Collections (ArrayList for books), custom exceptions,
// Comparator for sorting books
```

Week 6: File Handling & Multithreading Basics (Days 36-42)

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 36	File I/O (FileInputStream, FileOutputStream, Readers/Writers)	2	1.5	1.5	File operations, byte vs character streams
Day 37	Serialization (ObjectInputStream, ObjectOutputStream)	2	1	1.5	Object persistence, transient keyword
Day 38	Multithreading Fundamentals (Thread class, Runnable interface)	2.5	1	1.5	Thread creation, lifecycle, scheduling

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 39	Thread Methods (start(), run(), sleep(), join(), yield())	2	1.5	1.5	Thread synchronization basics
Day 40	Synchronization (synchronized keyword, locks, race conditions)	2	1.5	1.5	Thread safety, critical sections
Day 41	Practice Day - File operations & Multithreading	1	1.5	3	Concurrent file access, producer-consumer
Day 42	Review & Advanced Thread Concepts	1	1.5	2.5	Thread communication, deadlock prevention

Week 6 DSA Focus: Trees & Binary Search Trees

- Topics: Tree traversal (Inorder, Preorder, Postorder), BST properties, balancing
- Problem types: Invert tree, lowest common ancestor, level order traversal
- Target: 10-12 problems solved
- Complexity: Mix of Medium difficulty with some Hard problems

Week 6 Core Java Project: Build a **Multi-threaded File Processing System**

```
// Features: Multiple threads reading different files, concurrent file operations
// Classes: FileProcessor, SynchronizedFileHandler
// Demonstrates: Multithreading, synchronization, file I/O
// Example: Process multiple CSV files simultaneously, aggregate results
```

Week 7: Java 8+ Features & Functional Programming (Days 43-49)

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 43	Functional Interfaces (definition, @FunctionalInterface)	2	1	1.5	Single abstract method pattern
Day 44	Lambda Expressions (syntax, usage, function references)	2	1.5	1.5	Concise syntax, closures, variable capture
Day 45	Stream API Fundamentals (filter, map, reduce, collect)	2.5	1.5	1.5	Declarative programming, lazy evaluation
Day 46	Advanced Stream Operations (flatMap, groupingBy, partitioningBy)	2	1	1.5	Complex data transformations

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 47	Default Methods in Interfaces, Method References	2	1.5	1.5	Functional style programming
Day 48	Practice Day - Java 8 Features	1	1.5	3	Stream processing, functional composition
Day 49	Review & Functional Programming Paradigm	1	1.5	2.5	Performance implications of streams

Week 7 DSA Focus: Graphs & Graph Algorithms

- Topics: BFS, DFS, Dijkstra's algorithm, topological sort
- Problem types: Number of islands, course schedule, word ladder
- Target: 10-12 problems solved
- Important: Understand different graph representations (adjacency matrix vs list)

Week 7 Core Java Project: Build a **Data Processing Pipeline with Streams**

```
// Process large datasets using Stream API
// Features: Filter students by criteria, group by department, calculate statistics
// Demonstrates: Functional interfaces, lambda expressions, stream operations
// Example: Student records processing - filter by GPA, group by course,
// calculate average marks using streams
```

Week 8: Consolidation, DSA Problem Solving & Core Java Mastery (Days 50-56)

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 50	Review All Core Concepts (OOP, Collections, Streams)	2	2	2	Concept integration
Day 51	Advanced DSA - Dynamic Programming Intro (Memoization, Tabulation)	2	2.5	1.5	Optimization techniques
Day 52	DSA Problem Solving Marathon (Mixed difficulty problems)	1	3	2	Timed practice sessions
Day 53	Final Core Java Project Build	1	2	3	Comprehensive project combining all concepts
Day 54	DSA Contest Participation (LeetCode/HackerRank Weekly)	0.5	3.5	1.5	Competitive practice

Day	Topics	Learn (hrs)	DSA Practice (hrs)	Core Java Project (hrs)	Key Focus
Day 55	Core Java Interview Preparation & Common Issues	2	2	2	Interview Q&A, edge cases
Day 56	Final Review, Bug Fixes, Documentation	1	1.5	2.5	Code quality, clean code practices

Week 8 Milestones:

- **Total DSA problems solved:** 70-85 problems
- **Total Core Java projects:** 5-6 complete applications
- **Code quality:** All projects with proper error handling and documentation

Week 8 Comprehensive Project: Build a **Online Quiz Platform**

```
// Classes: User, Quiz, Question, Answer, Results
// Features: User registration, quiz creation, answer validation, score calculation, leac
// Demonstrates: All Phase 1 concepts - OOP, Collections, Streams, file I/O,
// exception handling, multithreading for report generation
```

PHASE 2: DATABASE & JDBC (Weeks 9-12 | Days 57-84)

Time Allocation Per Day: 8 Hours

- **2 hours**: Database theory & SQL learning
- **2 hours**: SQL practice & database design
- **2 hours**: JDBC implementation & projects
- **1.5 hours**: DSA intermediate problems
- **0.5 hours**: Review & documentation

Week 9: SQL Fundamentals & Database Design (Days 57-63)

Day	Topics	Learn (hrs)	SQL Practice (hrs)	JDBC Projects (hrs)	Key Focus
Day 57	Database Basics, RDBMS, Tables, Relationships	2	1.5	1	Normalization (1NF, 2NF, 3NF)
Day 58	Data Types, Constraints (PRIMARY, FOREIGN, UNIQUE, CHECK)	2	2	1	Database integrity
Day 59	SQL: SELECT, WHERE, Operators, Comparison	2	2	1	Query fundamentals

Day	Topics	Learn (hrs)	SQL Practice (hrs)	JDBC Projects (hrs)	Key Focus
Day 60	SQL: ORDER BY, LIMIT, DISTINCT, NULL handling	1.5	2.5	1	Data filtering & sorting
Day 61	SQL: JOIN types (INNER, LEFT, RIGHT, FULL, CROSS)	2.5	2.5	0.5	Complex queries, multiple tables
Day 62	SQL: Aggregate functions (SUM, AVG, COUNT, GROUP BY, HAVING)	2	2.5	0.5	Data summarization
Day 63	Practice Day & Database Schema Design	1	3	2	Design e-commerce/HR database schema

Week 9 DSA Focus: Continue with **Intermediate DSA**

- Topics: Recursion, dynamic programming (Fibonacci, coin change)
- Target: 8-10 problems with medium difficulty

Week 9 Database Project: Design & implement **E-Commerce Database**

```
Tables: Users, Products, Categories, Orders, OrderItems, Reviews, Payments
Relationships: Establish primary/foreign keys, indexes
Queries: Complex SELECT with JOINs, aggregations
```

Week 10: Advanced SQL & Query Optimization (Days 64-70)

Day	Topics	Learn (hrs)	SQL Practice (hrs)	JDBC Projects (hrs)	Key Focus
Day 64	Subqueries (scalar, row, table subqueries), IN/EXISTS	2	2.5	0.5	Query nesting, optimization
Day 65	Views, Stored Procedures, Functions	2	2	1	Database-level abstraction
Day 66	Indexing, Query Optimization, Execution Plans	2	1.5	1.5	Performance tuning
Day 67	Transactions, ACID properties, Locks, Rollback	2	1.5	1.5	Data consistency, concurrency
Day 68	Union, Intersect, Except, Case statements	2	2.5	0.5	Advanced query operations
Day 69	Practice Day - Complex SQL Problems	1	3	2	HackerRank SQL challenges
Day 70	Review & SQL Mastery	1	2	2	Interview preparation

Week 10 DSA Focus: Dynamic Programming & Optimization

- Topics: Fibonacci optimization, longest subsequence problems
- Target: 8-10 medium to hard problems

Week 10 Database Project: Implement **Banking System Database**

```
Tables: Accounts, Transactions, Users, Cards
Complex Queries: Account balance calculation, monthly statements,
fraud detection (unusual transactions)
Stored Procedures: Transfer funds, update balance (with transaction safety)
```

Week 11: JDBC & Database Connectivity (Days 71-77)

Day	Topics	Learn (hrs)	SQL Practice (hrs)	JDBC Code (hrs)	Key Focus
Day 71	JDBC Fundamentals (Connection, Statement, ResultSet)	2	0.5	2.5	Driver loading, connection pools
Day 72	PreparedStatement vs Statement, SQL Injection prevention	2	0.5	2.5	Security best practices
Day 73	ResultSet types (TYPE_SCROLL, CONCUR_UPDATABLE)	2	1	2	Advanced result set handling
Day 74	Exception handling in JDBC, Connection management	2	0.5	2.5	Proper resource closing (try-with-resources)
Day 75	Batch operations, Transactions in JDBC	2	1	2	Performance optimization
Day 76	Practice Day - JDBC applications	1	1	4	Build CRUD applications
Day 77	Connection Pooling, HikariCP basics	2	0.5	2.5	Connection efficiency

Week 11 DSA Focus: DP Problem Marathon

- Target: 12-15 medium difficulty DP problems
- Topics: Knapsack, longest palindrome substring, edit distance

Week 11 JDBC Project: Build a **Complete Student Management System with JDBC**

```
// Database: Students, Courses, Enrollments, Grades tables
// JDBC Operations:
// - CRUD operations with PreparedStatement
// - Get student transcript (complex query with JDBC)
// - Batch enrollment (batch operations)
```

```
// - Transaction support (atomic operations)
// - Error handling with custom exceptions
```

Week 12: Database Integration Consolidation (Days 78-84)

Day	Topics	Learn (hrs)	Practice (hrs)	Integration Projects (hrs)	Key Focus
Day 78	DAO Pattern (Data Access Objects)	2	1.5	2.5	Repository abstraction
Day 79	Data Transfer Objects (DTOs)	1.5	1.5	3	Layered architecture
Day 80	Practice & Optimization	1	2	3	Performance testing
Day 81	Multi-database support (MySQL, PostgreSQL)	1.5	1.5	3	Database portability
Day 82	Large-scale data handling	1	2	3	Bulk operations, pagination
Day 83	Integration Project Build	1	1.5	3.5	Full-stack application
Day 84	Review, Testing, Documentation	1	2	3	Code quality, readiness check

Phase 2 Final Project: Build a **HR Management System**

```
// Database: Employees, Departments, Salaries, Attendance, Performance
// Features:
// - Employee CRUD with validation
// - Department hierarchy
// - Salary calculation with deductions
// - Attendance tracking
// - Performance reviews with historical data
// - Complex reports (department-wise salary, top performers)
// Architecture: DAO pattern, DTOs, proper exception handling, logging
```

Phase 2 Milestones:

- SQL problems solved: 30+
- JDBC projects completed: 3-4
- Total DSA problems solved: 100+

PHASE 3: SPRING FRAMEWORK & SPRING BOOT (Weeks 13-20 | Days 85-140)

Time Allocation Per Day: 8 Hours

- **2-2.5 hours**: Spring Boot theory & concepts
- **2.5-3 hours**: Spring Boot hands-on coding
- **1.5 hours**: Integration projects
- **1-1.5 hours**: DSA advanced problems
- **0.5 hours**: Review & documentation

Week 13: Spring Framework Fundamentals (Days 85-91)

Day	Topics	Learn (hrs)	Core Concepts Practice (hrs)	Projects (hrs)	Key Focus
Day 85	Spring Framework overview, IoC, Dependency Injection	2.5	1.5	2	Inversion of Control principle
Day 86	Bean configuration (XML, Annotation, Java Config)	2	1.5	2.5	Different bean definition approaches
Day 87	Spring Annotations (@Component, @Autowired, @Qualifier)	2.5	2	2	Annotation-driven development
Day 88	Lifecycle of a Bean, Scope (Singleton, Prototype, Request)	2	1.5	2.5	Bean lifecycle management
Day 89	Constructor & Setter Injection, Circular Dependencies	2	1.5	2.5	Dependency injection patterns
Day 90	Practice Day - Spring Core Projects	1.5	2	3	Build simple spring applications
Day 91	Review & Spring Core Mastery	1.5	1.5	2.5	Interview preparation on Spring

Week 13 DSA Focus: Advanced Problem Solving

- Target: 12-15 hard difficulty problems
- Topics: Graphs, complex DP, backtracking

Week 13 Spring Project: Build a **Simple DI Container Application**

```
// Concepts: Manual bean creation, DI implementation
// Features: Create beans, inject dependencies, manage lifecycle
// Demonstrates: Understanding Spring's internal workings
```

Week 14: Spring Boot Fundamentals (Days 92-98)

Day	Topics	Learn (hrs)	Hands-on Coding (hrs)	Projects (hrs)	Key Focus
Day 92	Spring Boot intro, Auto-configuration, Embedded Tomcat	2	2	2	Rapid development setup
Day 93	Application Properties, Profiles (dev, test, prod)	2	2	2	Configuration management
Day 94	Spring Boot Starter POMs, Dependencies management	2	1.5	2.5	Maven/Gradle basics
Day 95	Actuator, Endpoints, Health monitoring	2	2	2	Production readiness
Day 96	Embedded database (H2), Testing with Spring Boot	2	2	2	Testing configuration
Day 97	Practice Day - Spring Boot Projects	1.5	2	3.5	Build working applications
Day 98	Review & Spring Boot Setup Mastery	1.5	1.5	2.5	Project initialization patterns

Week 14 DSA Focus: Continue advanced problem solving

- Target: 12-15 problems combining multiple concepts

Week 14 Project: Build a **Spring Boot Todo API**

```
// Features: Create, read, update, delete todos
// Endpoints: /todos (GET/POST), /todos/{id} (GET/PUT/DELETE)
// Database: H2 in-memory database
// Demonstrates: Spring Boot setup, application properties, embedded database
```

Week 15: REST APIs & Spring Web (Days 99-105)

Day	Topics	Learn (hrs)	REST API Development (hrs)	Projects (hrs)	Key Focus
Day 99	REST principles, HTTP methods, Status codes	2	1.5	2.5	RESTful design principles
Day 100	@RestController, @RequestMapping, @GetMapping etc.	2.5	2	2	API endpoints creation
Day 101	Request/Response handling (@RequestBody, @ResponseBody)	2	2	2	Data serialization (JSON)

Day	Topics	Learn (hrs)	REST API Development (hrs)	Projects (hrs)	Key Focus
Day 102	PathVariable, RequestParam, Headers handling	2	2	2	Advanced request mapping
Day 103	Response Entity, Custom responses, HTTP status codes	2	2	2	Proper response handling
Day 104	Practice Day - REST API building	1.5	2.5	2.5	Multiple complex APIs
Day 105	Review & REST API Best Practices	1.5	1.5	2.5	API design patterns

Week 15 DSA Focus: Practice timed coding sessions

- Target: 10-12 problems in time-bound manner (45 mins per problem)

Week 15 Project: Build a **Product Catalog REST API**

```
// Endpoints:
// GET /api/products - List all products (with pagination)
// GET /api/products/{id} - Get product by ID
// POST /api/products - Create new product
// PUT /api/products/{id} - Update product
// DELETE /api/products/{id} - Delete product
// GET /api/products/search?name=xyz - Search products
// Features: Proper error responses, status codes, JSON serialization
```

Week 16: Data Access with Spring Data JPA & Hibernate (Days 106-112)

Day	Topics	Learn (hrs)	ORM Development (hrs)	Projects (hrs)	Key Focus
Day 106	Hibernate overview, Entity mapping, @Entity, @Table	2.5	1.5	2	ORM fundamentals
Day 107	Relationships (@OneToOne, @OneToMany, @ManyToMany)	2.5	2	1.5	Complex entity relationships
Day 108	Spring Data JPA (Repositories, Queries, JPQL)	2	2	2	Abstraction over raw SQL
Day 109	Custom queries, Named queries, Paging/Sorting	2	2	2	Advanced data retrieval
Day 110	Lazy vs Eager loading, N+1 problem, Join fetch	2.5	2	1.5	Performance optimization
Day 111	Practice Day - Data persistence projects	1.5	2.5	2.5	Complex queries, relationships

Day	Topics	Learn (hrs)	ORM Development (hrs)	Projects (hrs)	Key Focus
Day 112	Review & Hibernate Best Practices	1.5	1.5	2.5	Interview questions on JPA

Week 16 DSA Focus: Revision and contest participation

- Target: 10 mixed difficulty problems + 1 contest

Week 16 Project: Build a **Blog Management System**

```
// Entities: User, Post, Comment, Category, Tag
// Relationships: User-Post (One-to-Many), Post-Comment (One-to-Many),
// Post-Category (Many-to-One), Post-Tag (Many-to-Many)
// Features:
// - CRUD for posts with Hibernate
// - Query user's posts with custom repository methods
// - Get all comments for a post (with optimization)
// - Search posts by category/tag
// - Pagination support
```

Week 17: Exception Handling & Validation (Days 113-119)

Day	Topics	Learn (hrs)	Implementation (hrs)	Projects (hrs)	Key Focus
Day 113	Global exception handling (@ControllerAdvice, @ExceptionHandler)	2	2	2	Centralized error handling
Day 114	Custom exceptions, Error response format	2	2	2	Consistent error responses
Day 115	Bean Validation (@Valid, @Validated annotations)	2	2	2	Input validation
Day 116	Custom validators, Validation groups	2	2	2	Complex validation logic
Day 117	Logging best practices (SLF4J, Logback)	2	2	2	Debugging & monitoring
Day 118	Practice Day - Error handling & validation	1.5	2.5	2.5	Robust API development
Day 119	Review & Production-ready APIs	1.5	1.5	2.5	Enterprise development patterns

Week 17 DSA Focus: Mock interviews with DSA problems

- Target: 8-10 problems + 1 technical interview simulation

Week 17 Project: Rebuild **Previous Projects with Proper Validation & Error Handling**

```
// Enhanced E-Commerce API:
// - Input validation (@Valid on all request bodies)
// - Global exception handler with proper error responses
// - Custom exceptions for business logic errors
// - Proper logging at all layers
// - Consistent JSON error response format
// Example error response:
// {
//   "timestamp": "2025-12-09T...",
//   "status": 400,
//   "error": "VALIDATION_FAILED",
//   "message": "Email format is invalid",
//   "fieldErrors": {...}
// }
```

Week 18: Security Fundamentals (Days 120-126)

Day	Topics	Learn (hrs)	Implementation (hrs)	Projects (hrs)	Key Focus
Day 120	Spring Security overview, Authentication vs Authorization	2.5	1	2.5	Security concepts
Day 121	User authentication, Password encoding (BCrypt)	2	2	2	Secure authentication
Day 122	JWT (JSON Web Token) basics, token generation & validation	2.5	2	1.5	Token-based auth
Day 123	JWT implementation in Spring Boot	2	2.5	1.5	Stateless authentication
Day 124	Authorization, Role-based access control (RBAC)	2	2	2	Fine-grained permissions
Day 125	Practice Day - Secure API development	1.5	2.5	2.5	Authentication/authorization flows
Day 126	Review & Security Best Practices	1.5	1.5	2.5	Secure coding practices

Week 18 DSA Focus: Continue contest participation

- Target: 1-2 online coding contests + 10 problems

Week 18 Project: Build a **Secure User Management API**

```
// Features:
// - User registration with password encryption
```

```
// - Login endpoint returning JWT token
// - Protected endpoints requiring JWT token
// - Role-based access (@PreAuthorize annotations)
// - Token refresh mechanism
// - Logout functionality
// Authentication flow: Register → Login → Get JWT → Use JWT for requests
```

Week 19: Advanced Spring Boot Features (Days 127-133)

Day	Topics	Learn (hrs)	Implementation (hrs)	Projects (hrs)	Key Focus
Day 127	Testing (Unit tests with JUnit, Mockito)	2	2	2	Testable code patterns
Day 128	Integration testing, @SpringBootTest	2	2	2	Full application testing
Day 129	API Documentation (Swagger/OpenAPI, SpringFox)	2	2	2	API documentation
Day 130	Docker basics, Containerizing Spring Boot apps	2	2	2	Deployment readiness
Day 131	AOP (Aspect-Oriented Programming), @Aspect	2	2	2	Cross-cutting concerns
Day 132	Practice Day - Advanced features integration	1.5	2.5	2.5	Full-featured application
Day 133	Review & Production Deployment Readiness	1.5	1.5	2.5	Enterprise-ready applications

Week 19 DSA Focus: Final DSA preparation

- Target: 15-20 mixed problems + 1 contest + mock interviews

Week 19 Project: Build a **Complete E-Commerce Backend with Advanced Features**

```
// Features:
// - User authentication & JWT security
// - Product catalog with search & filtering
// - Shopping cart management
// - Order processing with status tracking
// - Payment integration (Stripe simulation)
// - Reviews & ratings
// - Proper error handling & validation
// - Comprehensive test coverage (70%+ code coverage)
// - Swagger API documentation
// - Docker containerization
```

Week 20: Full Integration & Interview Preparation (Days 134-140)

Day	Topics	Learn (hrs)	Code Review (hrs)	Integration (hrs)	Key Focus
Day 134	Code review best practices, refactoring	2	2	2	Code quality improvement
Day 135	Design patterns (Singleton, Factory, Builder)	2	2	2	Design pattern application
Day 136	Microservices concepts (for future learning context)	2	1.5	2.5	Architecture understanding
Day 137	Database optimization, indexing strategies	2	2	2	Performance tuning
Day 138	Final project completion & documentation	1.5	2	3.5	Production-ready code
Day 139	Mock interviews & technical discussions	1.5	2.5	2	Interview preparation
Day 140	Resume building, Portfolio finalization	1.5	2.5	2	Job application readiness

Week 20 Final Comprehensive Project: Build a **Complete Full-Stack Application**

```
// Backend: Spring Boot with all learned concepts
// Features:
// - REST APIs for complex business logic
// - Database with multiple entity relationships
// - JWT-based security with role management
// - Comprehensive input validation
// - Global exception handling with proper logging
// - Unit & integration tests (70%+ coverage)
// - Swagger documentation
// - Docker containerization
// - CI/CD ready code structure
// - Clean code with design patterns
// - Performance optimization
```

Phase 3 Milestones:

- Spring Boot projects completed: 5-6
- Total lines of code written: 10,000+
- Test coverage: 70%+ across projects
- API endpoints created: 50+
- Code quality: Enterprise-level standards

PHASE 4: FULL-STACK INTEGRATION & JOB PREPARATION (Weeks 21-24 | Days 141-168)

Time Allocation Per Day: 8 Hours

- **1 hour**: Learning advanced concepts or industry trends
- **3 hours**: Building integrated projects
- **2 hours**: Interview preparation (DSA, system design, behavioral)
- **1.5 hours**: Resume, portfolio, LinkedIn optimization
- **0.5 hours**: Job applications & networking

Week 21: Frontend Basics Integration (Days 141-147)

Day	Topics	Learn (hrs)	Frontend Dev (hrs)	Integration (hrs)	Key Focus
Day 141	HTML, CSS, JavaScript fundamentals	1.5	2	2.5	Web technologies overview
Day 142	React basics (Components, JSX, Props, State)	2	2	2	Modern frontend framework
Day 143	React hooks (useState, useEffect, useContext)	2	2	2	State management in React
Day 144	API integration (fetch, Axios, React Query)	1.5	2.5	2	Frontend-backend communication
Day 145	Form handling, validation in React	1.5	2.5	2	User input management
Day 146	Practice Day - Full-stack integration	1	2.5	3.5	Frontend-backend working together
Day 147	Review & Full-stack understanding	1	2	2.5	End-to-end application flow

Week 21 DSA Focus: Final DSA problem marathon

- Target: 20 mixed difficulty problems
- Focus: Interview-style questions

Week 21 Project: Build a **Student Management Full-Stack Application**

```
Backend: Spring Boot APIs for student CRUD, course management
Frontend: React UI for managing students, enrollment, grades
Integration: Frontend consuming backend APIs, form validation, authentication
```

Week 22: Interview Preparation - DSA & System Design (Days 148-154)

Day	Topics	Mock Interviews	System Design	Job Prep (hrs)	Key Focus
Day 148	Technical interview patterns (Array, String problems)	2	1	2	Problem-solving strategies
Day 149	DSA Mock interview 1 (5 problems in 3 hours)	3	1	1	Timed practice
Day 150	System design basics (API design, DB schema)	1.5	2	2.5	Scalability thinking
Day 151	DSA Mock interview 2 (5 problems in 3 hours)	3	1	1	Consistency building
Day 152	Behavioral interview preparation (STAR method)	1	1.5	3.5	Soft skills
Day 153	DSA Mock interview 3 (5 problems in 3 hours)	3	1	1	Final polish
Day 154	Review weak areas & final adjustments	1	2	2.5	Gap analysis

Week 22 Milestones:

- Complete 3 full mock interviews
- System design Q&A preparation
- Behavioral stories documented
- Resume finalized

Week 23: Portfolio Building & Job Applications (Days 155-161)

Day	Tasks	Portfolio (hrs)	Job Prep (hrs)	Networking (hrs)	Key Focus
Day 155	GitHub repository organization	2	2	2	Professional presence
Day 156	Project documentation (README, architecture diagrams)	2	2	2	Clear communication
Day 157	Resume optimization for ATS (Applicant Tracking System)	2	3	1	Resume best practices
Day 158	LinkedIn profile complete with projects & endorsements	1.5	2.5	2.5	Professional networking
Day 159	Start job applications to target companies	1	3	2	Application strategy

Day	Tasks	Portfolio (hrs)	Job Prep (hrs)	Networking (hrs)	Key Focus
Day 160	Customized cover letters, tailored applications	1.5	3	1.5	Personalization
Day 161	Interview scheduling, confirmation	1	2	3	Active job hunting

Week 23 Portfolio Checklist:

- 5-6 projects on GitHub with complete documentation
- Each project has: README, project structure, setup instructions, usage examples
- Personal portfolio website (optional but valuable)
- LinkedIn with 500+ connections (network building)
- Resume with ATS optimization

Week 24: Active Job Hunting & Final Preparation (Days 162-168)

Day	Focus	Interviews	Learning (hrs)	Networking (hrs)	Key Focus
Day 162	Interview 1 (Company A)	1-2 hours	2	1.5	Real interview experience
Day 163	Interview 2 (Company B)	1-2 hours	2	1.5	Confidence building
Day 164	Interview 3 (Company C)	1-2 hours	2	1.5	Multiple opportunities
Day 165	Follow-ups, feedback from interviews	1	2	2	Continuous improvement
Day 166	Interview 4-5 (Additional companies)	2-3 hours	1.5	1.5	More interviews
Day 167	Offer negotiations, clarifications	1.5	1.5	2	Closing deals
Day 168	Final decision & celebration	1	1	2	Job secured!

DAILY ROUTINE STRUCTURE (Sample 8-Hour Day During Different Phases)

Phase 1 & 2 Daily Schedule (Weeks 1-12):

```
06:00 AM - 08:00 AM: Concepts Learning (Videos/Documentation) [2 hrs]
08:00 AM - 10:30 AM: Core Java/SQL Coding [2.5 hrs]
10:30 AM - 10:45 AM: Break
10:45 AM - 12:15 PM: DSA Problem Solving [1.5 hrs]
12:15 PM - 01:15 PM: Lunch Break
01:15 PM - 03:15 PM: Project Development [2 hrs]
03:15 PM - 03:30 PM: Break/Tea
03:30 PM - 04:30 PM: Documentation & Debugging [1 hr]
Total: 8 hours focused study
```

Phase 3 Daily Schedule (Weeks 13-20):

```
06:00 AM - 08:30 AM: Spring Boot Concepts [2.5 hrs]
08:30 AM - 11:00 AM: API Development Hands-on [2.5 hrs]
11:00 AM - 11:15 AM: Break
11:15 AM - 12:45 PM: DSA Problem Solving [1.5 hrs]
12:45 PM - 01:45 PM: Lunch Break
01:45 PM - 03:45 PM: Integration Project Building [2 hrs]
03:45 PM - 04:00 PM: Break
04:00 PM - 04:30 PM: Code Review & Documentation [0.5 hrs]
Total: 8 hours focused study
```

Phase 4 Daily Schedule (Weeks 21-24):

```
06:00 AM - 07:00 AM: Interview Preparation/Learning [1 hr]
07:00 AM - 10:00 AM: Project Completion/Enhancement [3 hrs]
10:00 AM - 10:15 AM: Break
10:15 AM - 12:15 PM: DSA Mock Interview/Problem Solving [2 hrs]
12:15 PM - 01:15 PM: Lunch Break
01:15 PM - 03:15 PM: Portfolio/Resume/Job Applications [2 hrs]
03:15 PM - 03:30 PM: Break
03:30 PM - 04:30 PM: Networking/LinkedIn/Interviews [1 hr]
Total: 8 hours
```

WEEKLY DSA PROBLEM QUOTA & PROGRESSION

Phase	Weeks	Total Days	Weekly Quota	Problem Level	Total by Phase End
Phase 1	1-8	56	8-10/week	Easy → Easy-Medium	70 problems
Phase 2	9-12	28	8-10/week	Medium	35 problems
Phase 3	13-20	56	12-15/week	Medium → Hard	100 problems
Phase 4	21-24	28	15-20/week	Hard + Mixed	70 problems

Phase	Weeks	Total Days	Weekly Quota	Problem Level	Total by Phase End
Total DSA Problems Solved:					**275+ problems**

RECOMMENDED LEARNING RESOURCES

Paid Resources (Optional but Valuable):

- **Udemy courses:** Complete Java Development, Spring Boot Master Class
- **Scaler Academy:** DSA + interview prep
- **InterviewBit:** DSA focused preparation

Free Resources (Sufficient for Success):

- **Core Java:** GeeksforGeeks, Official Java documentation
- **DSA:** LeetCode (free problems), HackerRank, CodeChef
- **Spring Boot:** Official Spring Boot documentation, Baeldung
- **SQL:** SQLZoo, Mode Analytics SQL tutorial
- **Full-Stack:** YouTube channels (Telusko, Code With Durgesh, Amigoscode)

Practice Platforms:

- **LeetCode:** 1500+ problems, contests, companies tag filters
- **HackerRank:** 500+ problems, interview prep
- **CodeChef:** Competitive programming
- **GitHub:** Collaborate, learn from open source

SUCCESS METRICS & CHECKPOINTS

By Week 8 (End of Phase 1):

- ✅ 70-85 DSA problems solved
- ✅ 5-6 Core Java projects completed
- ✅ Complete understanding of OOP, Collections, Streams
- ✅ Code quality: Clean, documented, tested

By Week 12 (End of Phase 2):

- ✅ 100+ DSA problems solved
- ✅ SQL proficiency: Can write complex queries
- ✅ JDBC mastery: Can build database-backed applications
- ✅ 3-4 database projects with proper DAO pattern

By Week 20 (End of Phase 3):

- ✅ 200+ DSA problems solved (targeting 250+)
- ✅ 5-6 Spring Boot projects with production quality
- ✅ API design skills: Can design scalable APIs
- ✅ Security understanding: JWT, Spring Security, password hashing
- ✅ Testing skills: 70%+ code coverage in projects

By Week 24 (Job Ready):

- ✅ 275+ DSA problems solved
- ✅ Professional GitHub portfolio with 5-6 projects
- ✅ Completed 3+ mock interviews
- ✅ Job offers from target companies
- ✅ Ready for senior developer collaboration

INTERVIEW PREPARATION TOPICS

Technical Round (70% Weight):

1. **Data Structures:** Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Hash Tables
2. **Algorithms:** Sorting, Searching, DFS, BFS, Dijkstra, DP, Backtracking
3. **Java Core:** OOP, Collections, Streams, Multithreading, Exception handling
4. **Databases:** SQL, JDBC, JPA/Hibernate, Normalization
5. **Spring Boot:** DI, REST APIs, JPA, Security, Testing
6. **System Design:** API design, database schema, scalability

Behavioral Round (30% Weight):

1. Project experience storytelling (STAR method)
2. Problem-solving approach
3. Learning ability & adaptability
4. Team collaboration

5. Handling failures & learning from mistakes

KEY SUCCESS FACTORS FOR 6-MONTH TIMELINE

1. **Consistency:** 8 hours daily without breaks—this is non-negotiable
2. **Project-first approach:** 70% building, 30% learning (reverse of most people)
3. **Code daily:** Even weekends, maintain momentum
4. **Review regularly:** Weekly reviews to consolidate learning
5. **Mock interviews:** Simulate real interviews starting week 17
6. **Stay focused:** Ignore distractions, follow the roadmap strictly
7. **Build in public:** Share progress on GitHub, LinkedIn
8. **Seek feedback:** Code reviews, peer discussions, mentorship

SALARY EXPECTATIONS & JOB PROSPECTS (2025-26)

With this intensive roadmap completion:

- **Expected CTC:** ₹3.5-5 LPA in India (starting)
- **Companies:** TCS, Infosys, Wipro, HCL, Cognizant, 100+ startups
- **Roles:** Junior Java Developer, Backend Engineer (Entry-Level), Software Developer
- **Growth:** Potential for ₹6-8 LPA after 2 years with continuous learning

This comprehensive roadmap, if followed diligently with 8 hours daily commitment, positions you for guaranteed job placement as a Java developer within 6 months.[2] [3] [4] [1]

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1. https://nikhillearnhub.com/nikhillearnhub/en/tech/roadmaps/roadmap-java.html
2. https://www.designgurus.io/answers/detail/how-can-i-practice-dsa-daily
3. https://www.linkedin.com/pulse/complete-java-freshers-guide-landing-your-first-job-2026-only-8d6tc
4. https://www.scaler.com/blog/dsa-roadmap/
5. https://talentsprint.com/blog/roadmap-successful-java-career-fresher
6. https://www.ccbp.in/intensive/spring-boot-course
7. https://www.kodnest.com/blog/java-fresher-interview-roadmap-a-week-by-week-preparation-guide
8. https://www.geeksforgeeks.org/java/a-quick-guide-on-dsa-and-competitive-coding-in-java/
9. https://www.codingshuttle.com/courses/java-spring-boot-0-to-100
10. https://nareshit.com/blogs/where-to-learn-spring-boot-as-back-end-skills